MATH 401. Homework #2. Sadullaev L.

1.2. Problem

In this exercise, all sets are subsets of Universe $\mathcal{U}$. Use the distributive laws & De Morgan's Laws to show that:

a) $(A^c \cup B)^c = A \setminus B$.

*b) $A \cap (B^c \cap A)^c = A \cap B$

c) $A^c \cap (B \cap C) = (B \setminus A) \cup (C \setminus A)$

Solution.

(b) Let $A, B \subseteq \mathcal{U}$. (1)

$$A \cap (B^c \cap A)^c = A \cap ((B^c)^c \cup A^c) = \text{(De Morgan's Law)}$$

$$= A \cap (B \cup A^c) = \text{(Complement of Complement is itself)}$$

$$= (A \cap B) \cup \underbrace{(A \cap A^c)}_{\emptyset} = \text{(Distributive L.) (considering (1))}$$

$$= (A \cap B) \cup \emptyset = A \cap B. \quad \square$$

1.3. Problem 2.

Show that $\bigcap_{n \in \mathbb{N}} \left(-\frac{1}{n}, \frac{1}{n}\right) = \{0\}$.

Solution:

Standart way of subset inclusion for both sides:

$(\supseteq)$

It is vacous that $0 \in \left(-\frac{1}{n}, \frac{1}{n}\right)$ symmetric interval for every $n \in \mathbb{N}$. (1)

$(\subseteq)$

Take an arbitrary point $x \in \left(-\frac{1}{n}, \frac{1}{n}\right)$ $\forall n \in \mathbb{N}$.

WTS:

If $x \in \bigcap_{n \in \mathbb{N}} \left(-\frac{1}{n}, \frac{1}{n}\right)$, $x \in \{0\}$. $\Rightarrow$ (2)

$\Rightarrow$ It seemed to me not so obvious, so I'm going to try contrapositive way:

Contrapositive approach for (2) might be

$$x \notin \bigcap_{\forall n \in \mathbb{N}} \left(-\frac{1}{n}, \frac{1}{n}\right) \Longleftarrow x \notin \{0\} \quad (3)$$

$$x \notin \{0\} \Rightarrow x \in U \setminus \{0\} = \mathbb{R} \setminus \{0\}$$

(def. of complement, $U = \mathbb{R}$ in our case)

Let $\forall x \in \mathbb{R} \setminus \{0\}$ (4)

WTS: $x \notin \bigcap_{n \in \mathbb{N}} \left(-\frac{1}{n}, \frac{1}{n}\right)$. (5)

If $|x| > 1$.

It is obvious $x \notin \bigcap_{n \in \mathbb{N}} \left(-\frac{1}{n}, \frac{1}{n}\right)$ (6).

If $0 < |x| \leq 1$, then

$|x| \leq 1$ (7) $\Rightarrow$

$\frac{1}{|x|} \geq 1$. (8) $\Rightarrow$

$$\frac{1}{|x|} \geq 1 \qquad (8) \qquad \Rightarrow.$$

$$\left\lceil \frac{1}{|x|} \right\rceil \geq \frac{1}{|x|} \qquad (9)$$ adding 1 to both sides:

$$\left\lceil \frac{1}{|x|} \right\rceil + 1 \geq \frac{1}{|x|} + 1 > \frac{1}{|x|} \Rightarrow$$

$$\left\lceil \frac{1}{|x|} \right\rceil + 1 > \frac{1}{|x|} \qquad (10)$$

Now, I've reached (10) which can make my thinking easy.

Let $m = \left\lceil \frac{1}{|x|} \right\rceil + 1$, so $m \in \mathbb{N}$ &

$$m = \left\lceil \frac{1}{|x|} \right\rceil + 1 > \frac{1}{|x|} \Rightarrow m > \frac{1}{|x|} \quad (11)$$

$$m > \frac{1}{|x|} \quad (11) \quad \text{for } m = \left\lceil \frac{1}{|x|} \right\rceil + 1 \in \mathbb{N}$$

$$= \frac{1}{m} < |x| \Rightarrow x \notin \left(-\frac{1}{m}, \frac{1}{m}\right) \quad (12)$$

$|x| > \frac{1}{m}$ for natural $m = \left\lceil \frac{1}{|x|} \right\rceil + 1$

$x \notin \left(-\frac{1}{m}, \frac{1}{m}\right)$ for some $m \in \mathbb{N}$ (13)

$\Rightarrow x \notin \bigcap_{n \in \mathbb{N}} \left(-\frac{1}{n}, \frac{1}{n}\right)$ (14) as some $n \geq m$.

So, by (6) & (14)

$\forall x \in \mathbb{R} \setminus \{0\} \Rightarrow x \notin \bigcap_{n \in \mathbb{N}} \left(-\frac{1}{n}, \frac{1}{n}\right) \Rightarrow$

$x \notin \mathbb{R} \setminus \{0\} \Leftarrow x \in \bigcap_{n \in \mathbb{N}} \left(-\frac{1}{n}, \frac{1}{n}\right)$ Contrapositive step $\Rightarrow$

$\forall x \in \bigcap_{n \in \mathbb{N}} \left(-\frac{1}{n}, \frac{1}{n}\right) \Rightarrow x \notin \mathbb{R} \setminus \{0\} \Rightarrow$

$\forall x \in \bigcap_{n \in \mathbb{N}} \left(-\frac{1}{n}, \frac{1}{n}\right) \Rightarrow x \in \left(\mathbb{R} \setminus \{0\}\right)^c = \{0\} \Rightarrow$

$\bigcap_{n \in \mathbb{N}} \left(-\frac{1}{n}, \frac{1}{n}\right) \subseteq \{0\}$ (15)

By (1), (6) & (15), we conclude $\bigcap_{n \in \mathbb{N}} \left(-\frac{1}{n}, \frac{1}{n}\right) = \{0\}$!

1.4. Problem 7.

Let X, Y be two nonempty sets & consider a function $f: X \to Y$.

(a) Show that if $B \subseteq Y$, then $f(f^{-1}(B)) = B$

(b) Show that if $A \subseteq X$, then $f^{-1}(f(A)) \supseteq A$. Find an example where $f^{-1}(f(A)) \neq A$.

Solution:

(a) [Preamble].

If $B \subseteq Y$, then

WTS: $f(f^{-1}(B)) = B$.

($\subseteq$)

Let $\forall\, y \in f(f^{-1}(B))$ (1) $\Rightarrow$

$\exists\, x \in f^{-1}(B) \quad \ni \quad y = f(x)$ (2) since

$f(f^{-1}(B)) = \{f(a) \mid a \in f^{-1}(B)\}$ (def. of image)

$f^{-1}(B) = \{x \in X \mid f(x) \in B\}$ (def. of preim.)

So, by (2) for $x \in f^{-1}(B)$, $y = f(x) \Rightarrow y = f(x) \in B$

$\Rightarrow$

$y \in B \Rightarrow f(f^{-1}(B)) \subseteq B$ (3)

($\supseteq$)

Let $\forall y \in B \subseteq Y$, then

$f^{-1}(B) = \{x \in X \mid f(x) \in B\}$ (4)
(def of preim)

$\Rightarrow \exists x \in f^{-1}(B) \ni y = f(x) \Rightarrow$.

$\Rightarrow y = f(x) \in f(f^{-1}(B))$ for $x \in f^{-1}(B)$ (def of image)

$y \in f(f^{-1}(B)) \Rightarrow B \subseteq f(f^{-1}(B))$ (5)

By (3) & (5) $B = f(f^{-1}(B))$.

(b) Let $f: X \to Y$ &
Assume $A \subseteq X$

WTS: $f^{-1}(f(A)) \supseteq A$.

($\supseteq$)

(2)

Let $x \in A$

$f(A) = \{ f(a) \mid a \in A \}$ (6) (def. of image)

$\Rightarrow \forall x \in A \quad y = f(x) \in f(A)$ based on (6).

$\Rightarrow y = f(x) \in f(A) \Rightarrow$

$f^{-1}(f(A)) = \{ x \in X \mid f(x) \in f(A) \}$ (def of preim.)

$\Rightarrow \exists x \in f^{-1}(f(A)) \ni y = f(x) \in f(A) \Rightarrow$

$x \in f^{-1}(f(A))$. $\Rightarrow$ (7)

$A \subseteq f^{-1}(f(A))$

Example. for $A \neq f^{-1}(f(A))$

Let $f(x) = x^2$, $f: \mathbb{R} \to \mathbb{R}$ and

Let $A = \{ x \mid x \in \mathbb{Z}^+ \} \subseteq X = \mathbb{R}$ (8)

So, $A = \{1, 2, 3, 4, ..\}$ (9)

Considering $f(x) = x^2$

$f(A) = \{1, 4, 9, 16, ..\}$ (10)

Using the definition of $f^{-1}(f(A))$:

$f^{-1}(f(A)) = \{x \in X = \mathbb{R} \mid f(x) \in f(A)\} \Rightarrow$

$f^{-1}(f(A)) = \{x \in \mathbb{R} \mid x^2 \in \{1, 4, 9, 16, ..\}\} \Rightarrow$

$f^{-1}(f(A)) = \{\pm 1, \pm 2, \pm 3, \pm 4, ..\}$ (11)

We can see here $A \subset f^{-1}(f(A))$

Specifically $A \neq f^{-1}(f(A))$. $\square$

1.4. Problem 8.

In this problem f, g are functions

$f: X \to Y$ & $g: Y \to Z$.

(a) Show that if f, g are injective, so is $g \circ f$.

(b) Show that if f, g are surjective, so is $g \circ f$.

(c) Explain that if f, g are bijective, so is $g \circ f$, and show that $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$.

Solution.

(a) Let $f: X \to Y \Rightarrow f(x) = y$ &

$g: Y \to Z \Rightarrow g(y) = z$.

Assume $f(x)$ & $g(y)$ are injective.

$f(x)$ injective $\Rightarrow f(x_1) \neq f(x_2)$ whenever $x_1 \neq x_2$ (1)

$g(y)$ injective $\Rightarrow g(y_1) \neq g(y_2)$ whenever $y_1 \neq y_2$ (2)

By (1), when $x_1 \neq x_2 \Rightarrow y_1 = f(x_1) \neq f(x_2) = y_2$

Let $y_1 = f(x_1)$ & $y_2 = f(x_2)$ then, by (2)

$g(y_1) \neq g(y_2) \Rightarrow g(f(x_1)) \neq g(f(x_2)) \Rightarrow$

$x_1 \neq x_2 \Rightarrow f(x_1) \neq f(x_2) \Rightarrow g(f(x_1)) \neq g(f(x_2)) \Rightarrow$

$g \circ f(x) = g(f(x))$ is injective.

(b) [Preamble]

Assume f, g are surjective. (1)

WTS: $g \circ f$ is surjective too.

By (1), using the def. of surjective:

$\forall y \in Y$, $\exists$ at least one $x \in X \ni f(x) = y$ (2)

$\forall z \in Z$, $\exists$ at least one $y \in Y \ni g(y) = z$ (3)

Let's consider $g(f(x))$.

Let $z = g(f(x))$. Since $g(y)$ is itself surjective, $\exists$ at least one $y \in Y \ni$ $z = g(y)$ (4)

Now we know that we have at least one $y \in Y \ni z = g(y)$.
But for $y \in Y$, $\exists$ at least one $x \in X \ni$ $y = f(x)$ as f is itself surjective. So
$z = g(y) \Rightarrow \exists y \in Y \Rightarrow \exists$ at least one $x \in X$ $\ni y = f(x) \Rightarrow$
for $z \in Z$, $z = g(f(x))$, $\exists$ at least one $x \in X$. $\Rightarrow$
$g \circ f$ is surjective.

(c) [Preamble]

Assume f, g are bijective. $\Rightarrow$

f bijective $\Rightarrow \begin{cases} (1)\ f \text{ is injective} \Rightarrow f(x_1) \neq f(x_2),\ x_1 \neq x_2 \\ (2)\ f \text{ is surjective} \Rightarrow \text{for } y = f(x)\ \exists\, x \in X \end{cases} \Rightarrow$

$\Rightarrow$ for $\forall y_1 = f(x_1)$ & $\forall y_2 = f(x_2)$ $\ni y_1 \neq y_2$,

for sure $\exists$ exactly one $x_1 \in X$, and one $x_2 \in X$. (3)

g bijective $\Rightarrow \begin{cases} (4)\ g \text{ is injective} \Rightarrow g(y_1) \neq g(y_2), \text{ if } y_1 \neq y_2 \\ (5)\ g \text{ is surjective} \Rightarrow \text{for } z = g(y),\ \exists \text{ at least one } y \in Y \end{cases}$

So,

$x_1 \neq x_2 \Rightarrow f(x_1) \neq f(x_2)$ for unique x_1 & unique x_2. (6)

$\Rightarrow$ Let $y_1 = f(x_1)$, $y_2 = f(x_2) \Rightarrow y_1 \neq y_2 \Rightarrow$ (7)

$\begin{cases} g(y_1) \text{ is unique value in } Z \ \& \\ g(y_2) \text{ is unique value in } Z \end{cases} \Rightarrow$ (8)

$g(y_1) \neq g(y_2) \Rightarrow g(f(x_1)) \neq g(f(x_2))$ if $x_1 \neq x_2$ (9)

By (8) & (9) $g \circ f$ is bijective

1.3. Problem 7.

If X is nonempty set, and $\mathcal{A}$ is a family of subsets of X, we call $\mathcal{A}$ an algebra of sets if the following three properties satisfied:

(i) $\emptyset \in \mathcal{A}$.

(ii) If $A \in \mathcal{A}$, then $A^c \in \mathcal{A}$ $(A^c = X \setminus A)$.

(iii) If $A, B, \in \mathcal{A}$, the $(A \cup B) \in \mathcal{A}$.

Assume that $\mathcal{A}$ is algebra of sets, then

(a) Show that $X \in \mathcal{A}$

(b) Show that if $A_1 \dots A_n \in \mathcal{A}$ $\forall n \in \mathbb{N}$, then

$$(A_1 \cup A_2 \cup \dots \cup A_n) \in \mathcal{A}.$$

(c) Show that if $A_1 \dots A_n \in \mathcal{A}$ $\forall n \in \mathbb{N}$,

$$(A_1 \cap A_2 \cap \dots \cap A_n) \in \mathcal{A}$$

(Hint: Use b), prop. (iii), De Morgan's Law).

Solution.

Let X be nonempty set &

$$\mathcal{A} = \{ A \mid A \subseteq X \} \qquad (1)$$

Assume $\mathcal{A}$ is alebra of sets

(a) Show that $X \in \mathcal{A}$ (2)

since $\mathcal{A} = \{ A \mid A \subseteq X \}$ & $X \subseteq X \Rightarrow$

$X \in \mathcal{A}$.

(b) $A_1, A_2 .. A_n \in \mathcal{A} \quad \forall n \in \mathbb{N}$

$$A_1 \cup A_2 \cup A_3 \cup_{..} \cup A_{n-1} \cup A_n =$$

$$\underbrace{\underbrace{\Big(\ldots \underbrace{\Big(\underbrace{\big(\underbrace{\overset{\in \mathcal{A}}{A_1} \cup \overset{\in \mathcal{A}}{A_2}}_{\in \mathcal{A} \text{ based on prop(iii)}} \big) \cup \overset{\in \mathcal{A}}{A_3}}_{\in \mathcal{A}} \Big) \cup \overset{\in \mathcal{A}}{A_4}}_{\in \mathcal{A}} \Big) \cup \ldots \cup A_{n-1} \Big)}_{\in \mathcal{A}} \cup \overset{\in \mathcal{A}}{A_n}}_{\in \mathcal{A} \text{ by prop. (iii).}}$$

(c) If $A_1, A_2 \dots A_n \in \mathcal{A}$ (3)

by prop. (ii) $A_i^c = (X \setminus A_i) \in \mathcal{A} \quad \forall i = \overline{1,n}$.

By (b) we learned that

$$B = \left(A_1^c \cup A_2^c \cup \dots \cup A_n^c\right) \in \mathcal{A} \quad (4)$$

$\forall A_i^c \in \mathcal{A}$

if $B \in \mathcal{A}$, again by prop.(ii) $B^c \in \mathcal{A}$:

$$B^c = \left(A_1^c \cup A_2^c \cup \dots \cup A_n^c\right)^c \in \mathcal{A} \quad (5)$$

$$\Rightarrow (A_1^c)^c \cap (A_2^c)^c \cap \dots \cap (A_n^c)^c \in \mathcal{A} \quad \left(\begin{array}{c}\text{By} \\ \text{De Morgan's} \\ \text{Law}\end{array}\right)$$

$$\Rightarrow A_1 \cap A_2 \cap \dots \cap A_n \in \mathcal{A}. \quad (6)$$

$\square$